Aakash Premnath Subramanyam

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EDUCATION

Northeastern University, Khoury College of Computer Sciences — Boston, MA
M.S. Cybersecurity — *Sep 2024 – Dec 2026*

Anna University, Sri Venkateswara College of Engineering — Tamil Nadu, India
B.E. Computer Science and Engineering — *Sep 2018 – Jul 2022*

WORK EXPERIENCE

Navitas Life Sciences — Boston, MA
Cybersecurity Intern — Security Auditing and AI-Augmented SOC — *Feb 2026 – Aug 2026*

- Identified the root cause of a **90% false positive rate** across ∼**50,000 daily SOC alerts** (Wazuh rule level 6–12 band); scoped and led a 6-month security audit spanning **CrowdStrike Next Gen SIEM, FireEye, Microsoft Purview DLP, and Netskope** — mapping all rules to **MITRE ATT&CK** and defining end-to-end remediation scope.
- Delivered the first Purview DLP policy overhaul to IT Ops for production rollout by auditing detection rules across all four platforms — tightening conditions, building whitelists, correcting thresholds, and aligning every rule to **MITRE ATT&CK**.
- Expanded SOC automation capability at **$0 additional cost** by identifying **Falcon Fusion SOAR** within the existing CrowdStrike licence; adopted as the orchestration and automated response layer across both AI project tracks.
- Designed and deployed an ML alert-triage tool on a two-model architecture — a fine-tuned **SecureBERT 2.0 + LoRA** classifier for inline threat/benign scoring, paired asynchronously with a **Foundation-Sec-8B** LLM judge — targeting the rule level 6–12 band with **MITRE ATT&CK** context extracted from Wazuh rule fields.
- Delivered under strict life-sciences data governance (open-weight, on-premises, **$0 software cost**): fine-tuned on **32K+ labelled Wazuh alerts** with recall-optimized **F2** training, deployed on an **AWS Tesla T4 GPU** behind **Netskope ZTNA**, and surfaced through a **React** SOC dashboard with a **FastAPI** LLM explanation service and prompt-injection hardening.

Wipro — Hyderabad, India
Project Engineer — Azure, Microsoft Entra — *March 2022 – Jul 2024*

- Reduced enterprise attack surface by **35%** across **10,000+ users** by implementing least-privilege enforcement and periodic access reviews in **Microsoft Entra ID**.
- Improved system uptime by **85%** and cut licensing overhead by **20%** by resolving **200+ monthly authentication incidents** at a **98% resolution rate** through root-cause analysis and licence optimization.
- Achieved **90% security compliance** across the enterprise user base by leading the integration of **O365 security features** and conditional access policies.

PROJECTS

Cloud SIEM with Hybrid Machine Learning for Attack Detection | *Security Operations Project*

- Unified syslog, auth.log, auditd, and Zeek log ingestion into a single normalized SIEM pipeline by building an **Azure-based cyber range** that replicated enterprise attack paths.
- Detected lateral movement, suspicious authentication, and privilege misuse across **140M+ log events** using a hybrid ML pipeline combining **Isolation Forest** (anomaly) and **Random Forest** (supervised classification).
- Achieved real-time attack-path visibility mapped to the **MITRE ATT&CK** framework by designing SOC-style dashboards and alert rules.

Malware Analysis and Development Research | *Academic Project*

- Mapped attacker TTPs across evasion, persistence, and stealth techniques by building PoC malware in isolated VMs using C and Python, analysing system calls, registry modifications, and behavioral indicators.
- Strengthened malware detection and incident response playbooks by producing defensive recommendations grounded in observed behavioral patterns.

TECHNICAL SKILLS

Security Platforms: CrowdStrike Next Gen SIEM (LogScale), Falcon Fusion SOAR, Microsoft Purview DLP, Netskope, Trellix, Wazuh (OpenSearch), Splunk, Wireshark, Burp Suite, Metasploit, Nmap

Cloud & Infrastructure: AWS EC2 (GPU), Azure Security Center, Microsoft Entra ID, Microsoft 365 (Purview, Defender), Docker, On-premises GPU Inference, CUDA

AI / ML: PyTorch, HuggingFace Transformers, LoRA/PEFT, FastAPI, SecureBERT 2.0, Foundation-Sec-8B, React, Scikit-learn (Isolation Forest, Random Forest, K-Means)

Programming & Frameworks: Python, C/C++, Bash, Git | LogScale (LQL), Falcon Query Language (FQL) | MITRE ATT&CK, NIST CSF, GxP, GDPR

CERTIFICATIONS & COURSEWORK

Microsoft Certified: Azure Fundamentals | Microsoft Certified: Security, Compliance, and Identity Fundamentals | Certified SAFe 6 Agilist | SC-500: Microsoft Certified: Cloud and AI Security Engineer Associate (beta) *(In Progress)*

Relevant Coursework: Network Security, Digital Forensics, Secure Software Development, Penetration Testing, Ethical Hacking, Security Risk Management